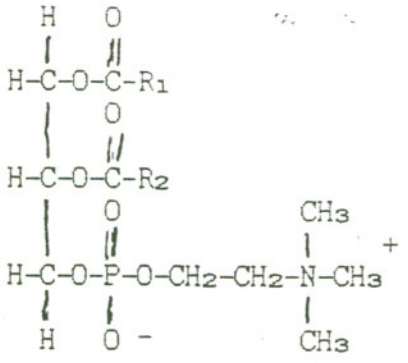
BIOCHEMISTRY CUMULATIVE EXAM
APRIL 27, 1991



Phospholipases are designated as phospholipase A if they catalyze the hydrolysis of either of the ester linkages between the fatty acids and glycerol, phospholipase C if they catalyze hydrolysis of the phosphoester linkage between gycerol and the phosphate, and D if they catalyze the hydrolysis of the bond between the phosphate and choline.

You have purified a phospholipase A. It migrates primarily as a single band on SDS gel electrophoresis with an apparent molecular weight of 25,000. It also elutes from a calibrated Bio-Gel P-150 column with an apparent molecular weight of 25,000. Ten ug of the enzyme in a "standard" one ml phospholipase A assay results in the release of 0.001 nMole of free fatty acid per minute from a phospholipid substrate. Ten ug of the same enzyme has no apparent activity with p-nitropheny phosphate or bis p-nitrophenyl phosphate (synthetic substrates for phophatase and diphosphatase assays). Likewise 10 ug of the enzyme does not catalyze any release of phosphate from AMP. Storage of the purified enzyme solution at 4°C for a week results in the formation of a fine precipitate. This precipitate was removed by filtration and found to be soluble in 37°C buffered saline. This preparation had no phospholipase activity (10 ug/ml assay), but produced the release of 0.01 uMole of Pi per minute (10 ug of enzyme/ml assay) from p-nitrophenyl phosphate and 0.008 uMole of Pi per minute (10 ug enzyme/ml assay) from AMP. No activity was detected with phosphatidyl choline or bis p-nitrophenyl phosphate (10 ug enzyme/ml assay). This preparation has an apparent molecular weight of 50,000 on the calibrated P-150 column. The precipitated protein comprises 2.2% of the total protein present in the original enzyme preparation.

(50 points) 1. You have proposed that the phospholipase monomer dimerizes to form an enzyme that has a different specificity, i.e. a phosphatase with no phospholipase activity. What two experiments would you conduct to prove or disprove your hypothesis? Describe these experiments in as much detail as time and knowledge allow.

(50 points) 2. Propose an alternate hypothesis to explain the results obtained. What experiments would confirm this second hypothesis? Again, describe in as much detail as time and knowledge allow.